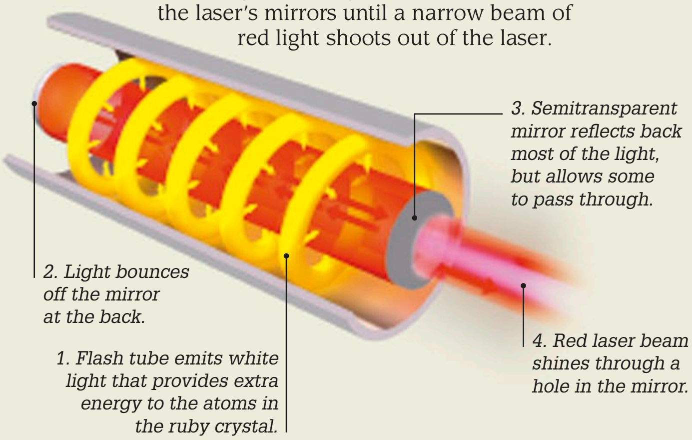
1960 ▸ 1965

1960 FIRST WORKING LASER

Dr. Theodore H. Maiman

Using a photographer's flash lamp and a ruby crystal rod, American engineer Theodore H. Maiman constructed a device that emitted a concentrated and focused beam of light known as a laser (Light Amplification by Stimulated Emission of Radiation). Lasers emit a single wavelength of light that stays focused and does not spread out, even when traveling long distances.

Maiman's ruby laser
Atoms within the ruby crystal rod are energized by the energy produced by the flash lamp firing. The atoms generate light that is reflected between the laser's mirrors until a narrow beam of red light shoots out of the laser.



Uses of lasers
Lasers are often used to measure distances in robotics and construction, while surgical lasers can seal blood vessels or destroy diseased cells. Industrial lasers can cut through steel and other tough materials (left) with unerring accuracy.

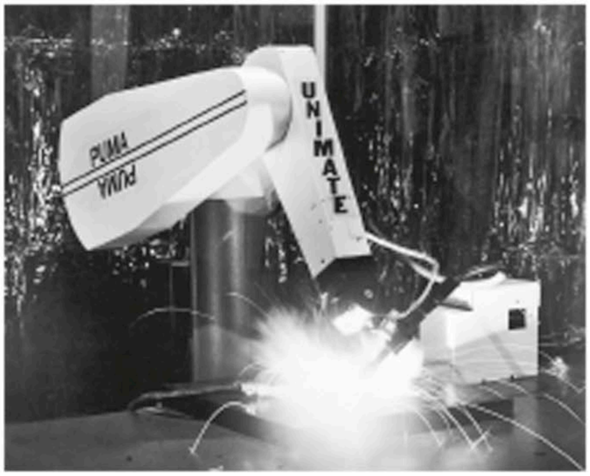
Trieste **is lowered into the Pacific Ocean**

1960 Deepest spot on Earth

US Navy Lieutenant Don Walsh and Swiss oceanographer Jacques Piccard descended to 35,797 ft (10,911 m) below sea level to Challenger Deep, the deepest part of the Pacific Ocean. Their vessel, a submersible called *Trieste*, had 5-in- (12.7-cm-) thick walls to withstand the immense water pressure at such depths—more than 1,080 times the pressure of Earth's atmosphere at sea level.

In 1960, *Tiros-1*, the first weather satellite, captured images of clouds and sent them to Earth via radio signals.

1960



Unimate 001 handling hot metal castings at a vehicle factory in New Jersey

1961 First industrial robot

A Unimate 001 robot arm was developed by American inventor George Devol and American physicist Joseph Engelberger. This 1.6-ton (1.5-metric-ton) hydraulics-powered robotic limb completed 100,000 hours of service by 1971.